

Complex Geometry

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§1 Complex numbers

On the first lecture we learned some basic information regarding complex numbers. There are no notes on this topic, but you can find all necessary information on the corresponding [Wikipedia page](#).

§2 Basics of complex geo

On this lecture we continued to discuss basic properties of complex numbers and got some basic formulas of geometric relations using complex numbers. These formulas can be easily found, for example in one of [Evan Chen's handouts](#).

§3 Complex circles

Consider a circle on a complex plane with center $\gamma \in \mathbb{C}$ and radius $R \in \mathbb{R}^+$. Its equation will be $|z - \gamma| = R$. We can rewrite it in the following way:

$$z\bar{z} - \bar{\gamma}z - \gamma\bar{z} + (\gamma\bar{\gamma} - R^2) = 0 \quad (1)$$

Note that $z\bar{z}$ and $\gamma\bar{\gamma} = R^2$ are real numbers, and $\bar{\gamma}z$, $\gamma\bar{z}$ are complex conjugates. Then we can consider a more general equation:

$$Az\bar{z} + Bz + C\bar{z} + D = 0; \quad A, D \in \mathbb{R}, \quad B, C \in \mathbb{C}, \quad C = \bar{B} \quad (2)$$

Definition 3.1. A matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is called a **Hermitian matrix** for the circle $\mathcal{C}$, which corresponds to the equation $Az\bar{z} + Bz + C\bar{z} + D = 0$

Every circle on $\mathbb{C}$ defines a Hermitian matrix. Note that matrices $\begin{pmatrix} \lambda A & \lambda B \\ \lambda C & \lambda D \end{pmatrix}$ define the same circle for $\forall \lambda \in \mathbb{R} : \lambda \neq 0$.

Definition 3.2. A circle is a Hermitian matrix such that $A \neq 0$ or $B \neq 0$.

Note that in this definition straight line (when $A = 0$) is also considered a circle.

Now we assume that $A \neq 0$. If a Hermitian matrix comes from a circle equation (1), we get:

$$\begin{aligned} Az\bar{z} - A\bar{\gamma}z - A\gamma\bar{z} + A(|\gamma|^2 - R^2) &= 0 \\ B &= -A\bar{\gamma}, \quad C = -A\gamma\bar{z} = \bar{B}, \quad D = A(|\gamma|^2 - R^2) \end{aligned} \quad (3)$$

$$\Delta(\mathcal{C}) = AD - BC = A^2(|\gamma|^2 - R^2) - A^2|\gamma|^2 = -A^2R^2 \in \mathbb{R}^- \quad (4)$$

Note that if $\Delta = 0$ the circle equation gives us a point $z_0 \in \mathbb{C}$.

Definition 3.3. Given Hermitian matrix for a circle $\mathcal{C}$, $A \neq 0$, we say that $\mathcal{C}$ is

- a **real circle**, if $\Delta(\mathcal{C}) < 0$
- a **point circle**, if $\Delta(\mathcal{C}) = 0$
- an **imaginary circle**, if $\Delta(\mathcal{C}) > 0$

We say that $\mathcal{C}$ is **positively oriented** if $A > 0$, and we say that $\mathcal{C}$ is **negatively oriented** if $A < 0$.

Now we can give classification for all circle equations corresponding to Hermitian matrices:

- 1.1 $A \neq 0, \Delta < 0$: real circle
- 1.2 $A \neq 0, \Delta = 0$: point circle
- 1.3 $A \neq 0, \Delta > 0$: imaginary circle
- 2.1 $A = 0, \Delta \neq 0$: line
- 2.2 $A = 0, \Delta = 0$: no equation

§4 Families of circles

Definition 4.1. Consider 2 circles $\mathcal{C}_1$ and $\mathcal{C}_2$, $\lambda\mathcal{C}_1 \neq \mathcal{C}_2$. Then the set $\{\lambda_1\mathcal{C}_1 + \lambda_2\mathcal{C}_2 \mid \lambda_{1,2} \in \mathbb{R}\}$ is called a **family of circles** or a **pencil**.

We can get that

$$\Delta(\mathcal{C} = \lambda_1\mathcal{C}_1 + \lambda_2\mathcal{C}_2) = \Delta_1\lambda_1^2 + 2\Delta_{12}\lambda_1\lambda_2 + \Delta_2\lambda_2^2, \quad (5)$$

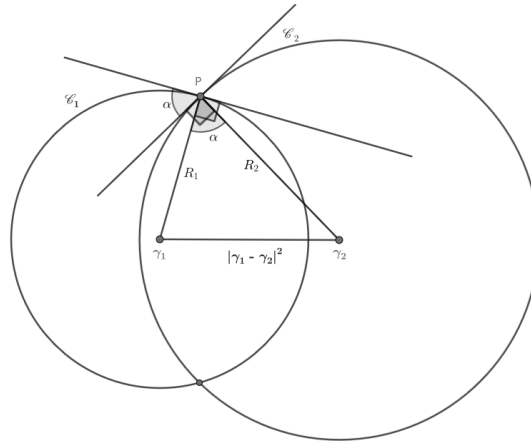
where $\Delta_{12} = A_1D_2 + A_2D_1 - (C_1B_2 + C_2B_1)$.

Note that $\Delta(\mathcal{C}) \in \mathbb{R}$ and it is a quadratic form in $\lambda_{1,2}$ with real coefficients.

Assume that $A_1, A_2 \neq 0$.

$$\Delta_{12} = A_1A_2(|\gamma_1 - \gamma_2|^2 - R_1^2 - R_2^2)$$

Assume $\mathcal{C}_1, \mathcal{C}_2$ are real circles ($\Delta < 0$). $\mathcal{C}_1, \mathcal{C}_2$ have a common point P :



From that we get the following equation (cosine theorem for $\Delta O_1 P O_2$):

$$|\gamma_1 - \gamma_2|^2 - R_1^2 - R_2^2 = \pm 2R_1R_2 \cos \alpha$$

And we get:

$$\Delta_{12} = \pm A_1A_2 \cdot 2R_1R_2 \cos \alpha$$

Therefore from the formula (4):

$$\Delta_1 = -A_1^2R_1^2 \Rightarrow \Delta_{12} = \pm \sqrt{-\Delta_1} \sqrt{-\Delta_2} \cos \alpha$$

Note that $|\cos \alpha| \leq 1$, so $\Delta_{12}^2 - \Delta_1\Delta_2 \leq 0$. Returning to equation (5), we get that the discriminant is non-positive, therefore every circle from a pencil of real circles is also real.

If equality holds ($\Delta_{12}^2 = \Delta_1\Delta_2$, $\cos \alpha = \pm 1$). We get:

$$|\gamma_1 - \gamma_2|^2 = (R_1 - R_2)^2$$

Definition 4.2. $\frac{-\Delta_{12}}{\sqrt{-\Delta_1}\sqrt{-\Delta_2}}$ is called the **common invariant** of $\mathcal{C}_1$ and $\mathcal{C}_2$, if $\mathcal{C}_1, \mathcal{C}_2$ are either both real or both imaginary circles.

Example 4.3

Suppose $\frac{-\Delta_{12}}{\sqrt{-\Delta_1}\sqrt{-\Delta_2}} > 1$ and $\mathcal{C}_1, \mathcal{C}_2$ are both real circles. Then we get that $(R_1 - R_2)^2 > \delta = |\gamma_1 - \gamma_2|^2$.

Similarly, if the common invariant < -1 , then $\mathcal{C}_1 \cap \mathcal{C}_2 = \emptyset$ and one of the circles is outside the other one.

Definition 4.4. Circles $\mathcal{C}_1$ and $\mathcal{C}_2$ are **orthogonal/perpendicular**, if $\Delta_{12} = 0$.

$$\Delta_{12} = 0 \iff \cos(\alpha) = 0 \iff \alpha = \frac{\pi}{2}.$$

Recall equation 5. We have several cases:

1. **Elliptic pencil** ($\Delta_{12}^2 - \Delta_1\Delta_2 < 0$). Then $\Delta_1, \Delta_2 < 0$. Also, $\mathcal{C}_1$ and $\mathcal{C}_2$ have 2 different common points. Moreover, every member of this family has the same intersection point and they are either a real circle or a point.
2. **Parabolic pencil** ($\Delta_{12}^2 - \Delta_1\Delta_2 = 0$). Assume $\mathcal{C}_1$ and $\mathcal{C}_2$ are real ($\Delta_i < 0$). Then $\Delta = -(\sqrt{-\Delta_1}\lambda_1 + \sqrt{-\Delta_2}\lambda_2)^2$. All the members are real and the pencil contains a point circle.
3. **Hyperbolic pencil** ($\Delta_{12}^2 - \Delta_1\Delta_2 > 0$). The pencil contains real and imaginary circles, as well as 2 point circles.

§5 Inversion

Consider the following question:

For a straight line l , there is a notion of "symmetry" with respect to l . Can we have some symmetry with respect to circles?

Proposition 5.1

Let $\mathcal{C}$ be a real circle and a point $z \in \mathbb{C}$, which lies neither on $\mathcal{C}$, nor at the center of $\mathcal{C}$. Then there is a unique point $z^* \neq z$, which is common to all circles perpendicular to $\mathcal{C}$ and through z .

Definition 5.2. z^* is called the symmetric point of z with respect to $\mathcal{C}$. This transformation is called **inversion** (with respect to circle $\mathcal{C}$). $\mathcal{C}$ is called the **fundamental circle** of the inversion.

Theorem 5.3

For every point z there is a unique inverse image.

Proof. Consider fundamental circle $\mathcal{C}_0$. We look for circles $\mathcal{C} \perp \mathcal{C}_0 \Rightarrow \Delta_{12} = 0$. Recall equations (2) and (5):

$$\begin{cases} \Delta_{12} = AD_0 + DA_0 - (BC_0 + B_0C) = 0 \\ Az\bar{z} + Bz + C\bar{z} + D = 0 \end{cases}$$

There are 3 independent variables (A, B, D) (because $C = \bar{B}$) and 2 equations, hence we have 2 independent solutions. Then there exist 2 circles $\mathcal{C}_1$ and $\mathcal{C}_2$, not proportional to each other. So we can find the pencil of these 2 circles. Since this is an elliptic pencil, every member has 2 common points: z and z^* . Therefore, such z^* exists for all z , moreover, this point is unique. $\square$

Proposition 5.4

We need to be able to compute z^* in terms of z and $\mathcal{C}_0$.

Again, we can make a system of linear equations:

$$\begin{cases} AD_0 + DA_0 - (BC_0 + B_0C) = 0 \\ Az\bar{z} + Bz + C\bar{z} + D = 0 \\ Az^*\bar{z}^* + Bz^* + C\bar{z}^* + D = 0 \end{cases}$$

We now have 4 complex variables (A, B, C, D) and 3 linear equations. Suppose all three equations are independent. Consider then tuples $(A, \operatorname{Re}(B), \operatorname{Im}(B), D) \subset \mathbb{R}^4$ which are in bijection with Hermitian tuples. Because the equations are independent, we have one free variable and 1 dimension of solutions. But there are 2 independent circles, therefore the space of solutions is at least 2-dimensional, so we have a contradiction. $z \neq z^* \Rightarrow$ equations 2 and 3 in the system are independent. Note that $(1, -\bar{z}, -z^*, \bar{z}z)$ is a solution to the last 2 equations of the system, so it must be a solution for the first one too.

Therefore we can substitute and get:

$$A_0 \bar{z} z^* + B_0 z^* + C_0 \bar{z} + D_0 = 0$$

$$z^* = -\frac{C_0 \bar{z} + D_0}{A_0 \bar{z} + B_0} \quad (6)$$

The problem is, $A_0 \bar{z} + B_0$ may be 0. To fix that, we need to state that $z \neq -\frac{\bar{B}_0}{A_0} = -\frac{C_0}{A_0}$. Note that $-\frac{C_0}{A_0}$ is a center of the circle $\mathcal{C}_0$. Previously, we had the assumption that point z is not the center of the circle, so we can say that the denominator is not 0.

Remark 5.5. When $\mathcal{C}_0$ is a straight line, z^* is precisely the point, symmetric to z with respect to the line $\mathcal{C}_0$.

Also note that if $z \in \mathcal{C}_0$, we get that $z^* = z$.

Therefore now we have a formula (6) for the inverse image of a point in complex coordinates.

Remember that previously we established that there are 2 independent solutions $\mathcal{C}_1$ and $\mathcal{C}_2$ - real circles (or points), such that $z \in \mathcal{C}_1 \cap \mathcal{C}_2$. Though if $\Delta_1 = 0$ or $\Delta_2 = 0$, at least one of them is a point circle. But then $\mathcal{C}_1$ meets $\mathcal{C}_0$ in z , but we assumed that $z \notin \mathcal{C}_0$, so we have a contradiction. Therefore, both circles are real and their pencil is elliptic.

Example 5.6

Suppose $\mathcal{C}_0$ is a unit circle ($|z| = 1$). From the formula (6), we get that $z^* = \frac{1}{\bar{z}}$. So if $z \in \mathcal{C}_0$, $z^* = z$ from the formula. Therefore, every point on the circle is an invariant with respect to inversion.

Let us find the image of an arbitrary point $w = (\frac{1}{2}, \frac{1}{2}) = \frac{1+i}{2}$: $w^* = 1 + i$. Note that w, w^* and 0 are collinear. In fact, for every point $w \in \mathbb{C} \neq 0$, its image z^* lies on the line through 0 and w . That can be checked the following way: $\frac{w^*}{w} = \frac{1}{\bar{w}} \cdot \frac{1}{w} = \frac{1}{|w|^2} \in \mathbb{R} \Rightarrow 0, w, w^*$ are collinear.

Proposition 5.7

Let $\mathcal{C}_0$ be a real circle with center γ_0 and radius $R_0 > 0$. Then w, w^*, γ_0 are collinear for $\forall w \neq \gamma_0$.

Proof. We checked the proposition for the unit circle. Note that we can make any circle a unit one, therefore this property works for an arbitrary real circle. $\square$

For a real circle $\mathcal{C}_0$ the inversion $f : z \mapsto z^*$ is a function from $\mathbb{C} \setminus \{\gamma_0\}$ to $\mathbb{C} \setminus \{\gamma_0\}$. Another property of inversion is that the inversion of a point is an **involution**, and therefore, a **bijection**.

Note that when z is close to γ_0 , z^* becomes very large. Then, to solve the problem with the center of the circle, we can introduce a new point by saying that $f(\gamma_0) = \infty$. To keep the properties of inversion, let $f(\infty) = \gamma_0$. If we look at the formula (6), the image of ∞ is defined properly. When $|z| \rightarrow \infty, z^* \rightarrow -\frac{C_0}{A_0} = \gamma_0$.

Definition 5.8. The complex plane $\mathbb{C}$ extended with a point ∞ is called the **extended complex plane**. Note that it is, in fact, a $\mathbb{CP}^1$.

Fix a non-point circle $\mathcal{C}_0$. There are 2 types of inversions:

- When $\mathcal{C}_0$ is real (**elliptic inversion**). We have the formula (6) and everything can be referenced to γ_0 , R_0 . We can get the following formula: $z^* = \gamma_0 + \frac{R_0^2}{\bar{z} - \gamma_0}$.
- When $\mathcal{C}_0$ is imaginary (**hyperbolic inversion**), denote symmetric point of z with z^{**} . The center and radius are $\gamma_0, i\rho_0 = R_0$, $\rho_0 \in \mathbb{R}$. Then we have formula $z^{**} = \gamma_0 - \frac{\rho_0^2}{\bar{z} - \gamma_0} = 2\gamma_0 - z^*$. So we can think about hyperbolic inversion as a composition of elliptic inversion and a point reflection.

Proposition 5.9

Every inversion carries circles into circles (in Hermitian matrix definition), real circles and straight lines to themselves, imaginary circles to imaginary circles.

Proof. From the definition of inversion, it suffices to check this for the unit circle $\mathcal{C}_0$. Remember the equation (2) of a circle. If $\{0\} \notin \mathcal{C}$, we get:

$$A + B \cdot \frac{1}{\bar{z}} + C \cdot \frac{1}{z} + D \cdot \frac{1}{z\bar{z}} = 0$$

$$A + Bz^* + C\bar{z}^* + Dz^*\bar{z}^* = 0$$

We can see that we got the equation of a circle again. If $\{0\} \in \mathcal{C}$, $D = 0$, and $f(\mathcal{C})$ is a straight line. Note that the determinant of the Hermitian matrix of a circle does not change after inversion.

Because hyperbolic inversion is an elliptic inversion composed with symmetry, hyperbolic inversion also has the property above. $\square$

Definition 5.10. Consider function f on the extended projective plane. f is **isogonal** at z_0 , if f carries two curves meeting at z_0 , making an angle Θ , to two curves meeting in $f(z_0)$, making an angle $\Omega = \pm\Theta$. f is **conformal**, if $\Omega = \Theta$

Proposition 5.11

Inversions are isogonal.

Lemma 5.12

Consider an inversion with fundamental circle $\mathcal{C}_0, (O, r)$. Then for every 2 points A, B :

$$\overline{OA} \cdot \overline{OA^*} = r^2 \tag{7}$$

$$|A^*B^*| = |AB| \cdot \frac{r^2}{|OA| \cdot |OB|} \tag{8}$$

Example 5.13 (Ptolemy's Theorem)

Ptolemy's theorem says that for non-collinear cocyclic points A, B, C, D , which lie on a circle ω in that order, the following is true:

$$|AD| \cdot |BC| = |AC| \cdot |BD| + |AB| \cdot |CD|$$

Proof. To prove that, consider an inversion in A with an arbitrary radius. After using the formula (8), we need to prove that $|B^*C^*| + |C^*D^*| = |B^*D^*|$, which is obvious, because they lie on the straight line after the inversion in order $C^* \in [B^*D^*]$. $\square$

Example 5.14 (Euler's Circle)

Let ABC be a triangle with circumcenter O , circumradius R , and incenter I and inradius r . Then the following formula is true: $\overline{OI} = R^2 - 2Rr$.

Proof. Consider circle XYZ , where X, Y, Z are tangent points of the incircle and the initial triangle. Suppose A', B', C' are midpoints of the sides of this triangle, and consider $(A'B'C')$, which is the Euler's circle for this triangle. Remember that the radius of the nine-point circle is half the circumradius of the triangle. Now, note that

$$\Delta CC'Y \sim \Delta CYI \Rightarrow \overline{CI} \cdot \overline{C'I} = r^2.$$

Similarly:

$$\overline{AI} \cdot \overline{A'I} = \overline{BI} \cdot \overline{B'I} = r^2$$

Then we get that $\mathcal{C}(ABC)$ is the image of $\mathcal{C}(A'B'C')$ under the inversion with $\mathcal{C}(XYZ)$. Now we have that:

- $\mathcal{C}_0$ - fundamental circle (XYZ) , $|z|^2 = \rho^2 > 0$
 - $\mathcal{C}_2$ - $(A'B'C')$ - real circle
 - $\mathcal{C}_1$ - (ABC) , image of $\mathcal{C}_2$ after the inversion in $\mathcal{C}_0$
- $$\mathcal{C}_1 : Az\bar{z} + Bz + C\bar{z} + D = 0$$

After the inversion (suppose $\mathcal{C}_0$ is the unit circle):

$$Dz^*\bar{z}^* + B\rho^2 z^* + C\rho^2 \bar{z}^* + A\rho^4 = 0$$

Recall that

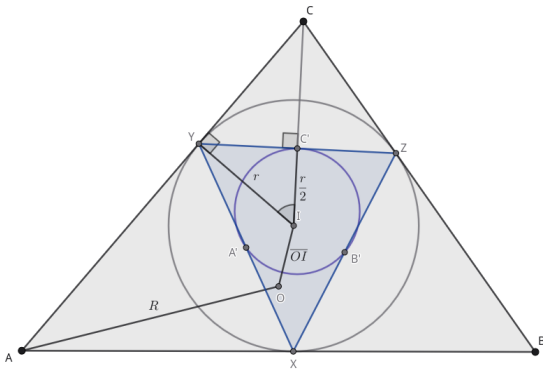
$$\Delta = -A^2 \cdot (\text{radius})^2 \Rightarrow \text{radius}_{\mathcal{C}_2} = \frac{1}{|A|} \cdot \sqrt{-\Delta}$$

$$\text{radius}_{\mathcal{C}_1} = \frac{1}{|D|} \sqrt{-\rho^4 \Delta} = \frac{1}{|D|} \rho^2 \sqrt{-\Delta}$$

Now let us compute the ratio of the radii of $\mathcal{C}_1$ and $\mathcal{C}_2$:

$$\frac{r}{R} = \frac{\frac{1}{|D|} \rho^2 \sqrt{-\Delta}}{\frac{1}{|A|} \sqrt{-\Delta}} = \frac{|A|}{|D|} \rho^2 = \frac{1}{R^2 - \overline{OI}^2} r^2$$

$\square$



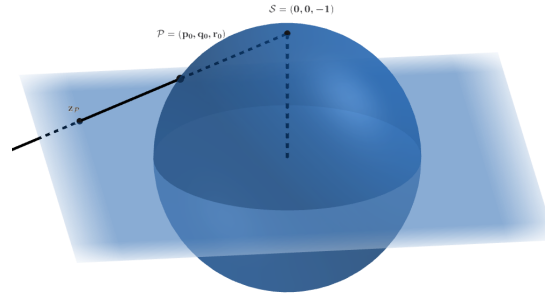
§6 Stereographic projections

Recall that for inversion to be a complete bijection we need to extend the complex plane $\mathbb{C}$ with one point at infinity to get $\mathbb{CP}^1$. This can be geometrically introduced with stereometric projection.

Definition 6.1. Consider $\mathbb{R}^3$ with coordinates $(p; q; r)$ and a sphere $S^2 = \{(p, q, r) \mid p^2 + q^2 + r^2 = 1\}$.

$$\begin{aligned}\overline{\mathcal{PS}} &= \{\lambda \mathcal{P} + (1 - \lambda)S \mid \lambda \in \mathbb{R}\} \\ &= \{\lambda(p_0, q_0, r_0) + (1 - \lambda)(0, 0, -1) \mid \lambda \in \mathbb{R}\} \\ \lambda \cdot r_0 + \lambda &= 1 \Rightarrow \lambda = \frac{1}{1 + r_0} \\ z_{\mathcal{P}} &= \left(\frac{p_0}{1 + r_0}, \frac{q_0}{1 + r_0}, 0\right) = \frac{p_0 + q_0 i}{1 + r_0} \in \mathbb{C}\end{aligned}$$

$f : S^2 \setminus \{S\} \rightarrow \mathbb{C}$ is called **stereographic projection**. Note that f is bijective.



Now the stereographic projection allows us to define $\overline{\mathbb{C}} = \mathbb{C} \cup \infty = (S^2 \setminus \{S\}) \cup \{S\}$. Also note that this function is continuous.

Proposition 6.2

The stereographic projection carries circles on $\mathbb{C}$ to circles on S^2 . A circle in S^2 is an affine hyperplane in $\mathbb{R}^3$.

Moreover, it takes real circles (straight lines) to real circles (hyperplanes which intersect S^2 non-trivially), it takes imaginary circles to imaginary circles (hyperplanes which do not intersect S^2).

Proof. If we have a Hermitian matrix of a circle $\mathcal{C}$, we can apply stereographic projection and get the equation $ap + bq + cr + d = 0 \mid p, q, r \in \mathbb{R}$, where:

$$a = B + C, \quad b = i(B - C), \quad c = A - D, \quad d = A + D \quad (9)$$

The distance between the hyperplane $ap + bq + cr + d = 0$ and the point $(0, 0, 0)$ is equal to $\frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$.

$$\begin{aligned}a^2 + b^2 + c^2 &= (B + C)^2 - (B - C)^2 + (A - D)^2 = 4BC + (A - D)^2; \quad |d|^2 = (A + D)^2 \\ dist &= \frac{(A + D)^2}{4BC + (A - D)^2} \geq 1 \Leftrightarrow (A + D)^2 \geq 4BC + (A - D)^2 \Leftrightarrow AD - BC = \Delta(\mathcal{C}) \geq 0\end{aligned}$$

Therefore, if $dist > 1$, the hyperplane does not touch the sphere (imaginary hyperplane), and it corresponds to the circle with $\Delta > 0$ - imaginary circle. Similarly, the other types of planes correspond to their types of circles. $\square$

Conclusion:

- $\Delta < 0$ (real circle) $\leftrightarrow$ hyperplane intersects S^2 .
- $\Delta = 0$ (point circle) $\leftrightarrow$ hyperplane is tangent to S^2 .
- $\Delta > 0$ (imaginary circle) $\leftrightarrow$ hyperplane does not touch S^2 .

Definition 6.3. A **metric space** is a set X together with a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$, which is called a **metric**, which satisfies the following conditions:

- $d(x, y) = 0 \Leftrightarrow x = y$.
- $d(x, y) = d(y, x)$, $\forall x, y$.
- $d(x, y) + d(x, z) \geq d(y, z)$, $\forall x, y, z$.

Example 6.4 (Metric Spaces)

Some examples of metric spaces:

1. $X = \mathbb{C}$, $d(z, w) = |z - w| = (z - w)(\overline{z - w})$.
2. $X = \mathbb{R}^3$, $d(x, y) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$.
3. $X = \mathbb{R}$, $d(p, q) = (p == q)$ (discrete metric).

Definition 6.5. Let $(X, d_X), (Y, d_Y)$ be metric spaces. A function (map) $f : X \rightarrow Y$ is called **continuous** at $x \in X$, if:

$$\begin{aligned} \forall \varepsilon > 0 \Rightarrow \exists \delta > 0 : \forall t : d_X(t - x) < \delta \Rightarrow d_Y(f(t) - f(x)) < \varepsilon \Leftrightarrow \\ \Leftrightarrow \forall \varepsilon > 0 \Rightarrow \exists \delta > 0 : \mathcal{B}_\delta(x) \subset f^{-1}(\mathcal{B}_\varepsilon(f(x))); \quad \mathcal{B}_\delta(x) = \{t \in X \mid d(t, x) < \delta\} \end{aligned}$$

Claim 6.6 — The stereographic projection is continuous $f : S^2 \setminus \{\mathcal{S}\} \rightarrow \mathbb{C}$.

Remark 6.7. There is a special kind of continuous functions. Suppose $f : (X, d_X) \rightarrow (Y, d_Y)$ is a continuous function. Then if $d_X(x, y) = d_Y(f(x), f(y))$, $\forall x, y \in X$, such function is called a **rigid motion**.

One of the projects proposed during this camp will be tied to this kind of functions.

We will say **extended inversion**, when we will be talking about inversion on the extended complex plane. Note that the extended inversion is a continuous function from S^2 to itself.

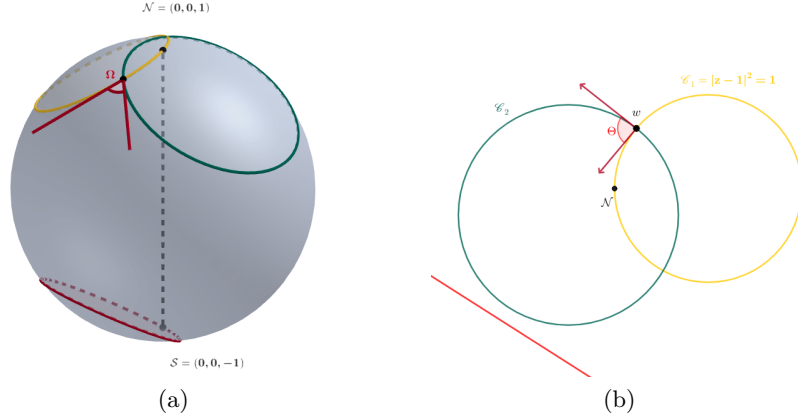
$S^2 \setminus \{\mathcal{S}\} \rightarrow \mathbb{C}, S^2 \setminus \{\mathcal{N}\} \rightarrow \mathbb{C}$. Note that this way, S^2 can be "covered" with 2 complex planes. Then S^2 has the following property: every point $p \in S^2$ has a neighborhood which looks like a piece of Euclidean plane. S^2 is referred to as **manifold**.

Now let us return to the map:

$$f : \mathbb{C} \rightarrow S^2 \setminus \{\mathcal{S}\}; \quad z \rightarrow \frac{1}{1 + |z|^2} (z + \bar{z}, -i(z - \bar{z}), 1 - |z|^2)$$

Proposition 6.8

Stereographic projection is a conformal map, i.e. it preserves angles.



Proof. Let us first understand how to compute "tangent vectors" of circles on S^2 . For that consider family of circles, generated by circles $|z - 1| = 1$ and $|z - w| = 0$. The we will find such a circle, that its hyperplane would pass through point $(0, 0, 0)$ in $\mathbb{R}^3$.

$$Az\bar{z} + Bz + C\bar{z} + D = 0; \quad Aw\bar{w} + Bw + C\bar{w} + D = 0$$

$$|z - w| = 0 \Rightarrow z\bar{z} - \bar{w}z - w\bar{z} + w\bar{w} = 0$$

$$\exists \lambda \in \mathbb{R} : (A + \lambda)z\bar{z} + (B - \lambda\bar{w})z + (C - \lambda w)\bar{z} + D + \lambda|w|^2 = 0; \quad A + D + \lambda + \lambda|w|^2 = 0$$

Then for this λ the circle will pass through the center of coordinates, moreover, it will have the same angle with the circle $\mathcal{C}_2$. So we can assume $A_i + D_i = 0$, $i = 1, 2$. Now let us calculate the angle between the 2 planes:

$$\cos \Omega = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}} = \cos \Theta (= \frac{\Delta_{12}}{\sqrt{-\Delta_1}\sqrt{-\Delta_2}})$$

If we directly compute these 2 values, we shall get that $\Omega = \Theta$. $\square$

Definition 6.9. Let Γ be a circle on S^2 (an affine hyperplane in $\mathbb{R}^3$). A point $Q \in S^2$ is called a **symmetric point** of $\mathcal{P} \in S^2$ with respect to Γ , if every circle (hyperplane), perpendicular to Γ through $\mathcal{P}$, also passes to Q .

Now let us understand how to find a set of symmetric points to $\mathcal{P}$:

$$\Gamma\{ap + bq + cr + d = 0\}; \quad \mathcal{P} = (p_0, q_0, r_0) \in S^2\}$$

An affine hyperplane $a'p + b'q + c'r + d' = 0$ is perpendicular to Γ , if (note also that because of conformality, the circles are perpendicular, therefore):

$$\begin{aligned} 0 &= aa' + bb' + cc' = (B + C)(B' + C') - (B - C)(B' - C') + (D - A)(D' - A') = \\ &= 2BC' + 2B'C + DD' - DA' - AD' + AA' = BC' + CB' + DD' + AA' = \\ &= AD' + A'D + DD' + AA' = (A + D)(A' + D') = d \cdot d' \end{aligned}$$

We have the following equations:

$$a'a + b'b + c'c - d'd = 0$$

$$a'p_0 + b'q_0 + c'r_0 - d' = 0$$

Suppose these 2 equations are linearly independent. Let Γ_1, Γ_2 be 2 independent solutions. Then $\Gamma_1 \cap \Gamma_2$ is a straight line in $\mathbb{R}^3$, which contains $\mathcal{P}$. Then the symmetric point of $\mathcal{P}$ is $Q = S^2 \cap (\Gamma_1 \cap \Gamma_2) \neq \mathcal{P}$.

We have another case: 2 equations are linearly dependent. Then Γ , the only solution is tangent of S^2 at $\mathcal{P}$ and we get that $\mathcal{P}$ is symmetric to itself.

Definition 6.10. Suppose $\mathcal{P} = (p_0, q_0, r_0) \in \mathbb{R}^3$. Then the affine hyperplane $p_0p + q_0q + r_0r = 1$ is called the **polar plane** of point $\mathcal{P}$, and $\mathcal{P}$ is called the **pole** of that plane.

Remark 6.11. If $\mathcal{P} \in S^2$, then the polar plane of $\mathcal{P}$ is tangent to the sphere in $\mathcal{P}$.

Theorem 6.12

Let $\mathcal{P} \in \mathbb{R}^3$, $\mathcal{P} \neq (0, 0, 0)$. Then $\forall Q \in \alpha$, where α is a polar plane of $\mathcal{P}$, polar plane of Q must contain $\mathcal{P}$. Also, the pole of plain passing through $\mathcal{P}$, but not the origin, lies in the polar plane of $\mathcal{P}$.

Proof. $Q = (a, b, c) \in \mathbb{R}^3 : p_0a + q_0b + c_0r = 1$. Then the polar plane of Q is $ap + bq + cr = 1 \Rightarrow \mathcal{P}$ lies on the polar plane of Q . $\square$

Now we consider the geometric interpretation:

$\mathcal{P} = (p_0, q_0, r_0) \in \mathbb{R}^3$, $\mathcal{P}$ is outside the S^2 . Let Γ be the intersection of the polar plane of $\mathcal{P}$ and the sphere. Then for $\forall Q \in \Gamma$, the polar plane of Q will be the tangent plane to S^2 in Q and contain $\mathcal{P}$. Then to compute the polar plane of $\mathcal{P}$ we can consider a circle of points, such that their tangent planes contain $\mathcal{P}$ and the hyperplane cutting it will be the polar plane.

If $\mathcal{P}$ lies inside S^2 , its polar plane does not touch S^2 . To find it, we consider 3 different hyperplanes through it and we find their poles. Then with these 3 points we can construct the plane, which will be polar plane for the point $\mathcal{P}$.

Proposition 6.13

Let Γ be a circle on S^2 , $P \in S^2 \setminus \Gamma$. Then the symmetric point P^* of P will be given by second intersection of S^2 and the line through P and the pole of the hyperplane determined by Γ .

§7 Cross-Ratio

Consider $\overline{\mathbb{C}}$. We need to carefully define addition and multiplication when working with infinity:

1. $a \neq \infty$, then we define $a \pm \infty := \infty$, $\frac{a}{\infty} := 0$.

2. $a \neq 0$, then we define $a \cdot \infty := \infty$, $\frac{a}{0} := \infty$.

Note that we don't define the results of $\frac{0}{0}, \frac{\infty}{\infty}, \infty \pm \infty, 0 \cdot \infty$.

Remark 7.1. $\infty \in \overline{\mathbb{C}}$ can be thought as a limit of sequence $\{z_n\}_{n=1}^{\infty}$, with $\lim_{n \rightarrow \infty} |z_n| = \infty$.

Definition 7.2. The sequence $\{z_n\}_{n=1}^{\infty}$ is called **convergent**, if $\exists \lim_{n \rightarrow \infty} z_n \in \mathbb{C}$.

Definition 7.3. Sequence $\{x_n\}_{n=1}^{\infty}$ is called **Cauchy sequence**, if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} : \forall n, m > N \Rightarrow |x_n - x_m| < \varepsilon.$$

In $\mathbb{C}$, the sequence is Cauchy iff it is convergent with a limit in $\mathbb{C}$.

So we can "define" a number on $\mathbb{C}$ as a limit of some Cauchy sequence.

Definition 7.4. We say that sequences $x = \{x_n\}_{n=1}^{\infty}$ and $y = \{y_n\}_{n=1}^{\infty}$ are **equivalent** ($\sim$), if:

$$\lim_{n \rightarrow \infty} |x_n - y_n| = 0.$$

Note that in this definition equivalence relation satisfies the equivalence conditions:

$$x \sim x; \quad x \sim y \Rightarrow y \sim x; \quad x \sim y, y \sim z \Rightarrow x \sim z.$$

We can then partition our sequences into several equivalence classes (classes where every 2 elements are equal in terms of that relation). We will denote equivalence classes as $[x]$, where $x \in [x]$. Note that these classes will be disjoint subsets of set of all Cauchy sequences, moreover, every sequence will be in some equivalence class.

Now we have a formal definition for ∞ point on $\overline{\mathbb{C}}$, as an equivalence class $[i]$ of all sequences $\{i_n\} : \lim_{i_n} \frac{1}{i_n} = 0$. We can define every other point z as class $[z]$ of sequence $z_n \equiv z$. Then we can check all properties of the infinity point, described in the beginning of the section. After establishing the nature of infinity point properties, let us come back to the $\overline{\mathbb{C}}$.

Definition 7.5. Suppose $z_1, z_2, z_3, z_4 \in \mathbb{C}$. Then **simple ratio** of points z_1, z_2, z_3 (with the assumption that not all of them are equal) is $(z_1; z_2, z_3) := \frac{z_1 - z_2}{z_1 - z_3}$. If $z_1 = z_3$, the value is equal to ∞ .

If $(z_1; z_2, z_3) = r$, then:

$$(z_2; z_3, z_1) = \frac{r-1}{r}; \quad (z_3; z_1, z_2) = \frac{1}{1-r}; \quad (z_1; z_3, z_2) = \frac{1}{r};$$

$$(z_2; z_1, z_3) = \frac{r}{r-1}; \quad (z_3; z_2, z_1) = 1-r.$$

We can extend the definition of a simple ratio on $\overline{\mathbb{C}}$ and allow (at most) one of the point to be ∞ :

$$(\infty; z_2, z_3) = 1, \quad (z_1; \infty, z_3) = \infty, \quad (z_1; z_2, \infty) = 0.$$

We need to pay attention to the following problem: if we have 2 points ∞ and calculating simple ratio, we cannot get a unique answer, therefore, we will not determine simple ratio for that case.

Definition 7.6. For $z_1, z_2, z_3, z_4 \in \mathbb{C}$, such that any three of them are not equal, consider 4 simple ratios:

$$(z_1; z_3, z_4), (z_2; z_3, z_4), (z_3; z_1, z_2), (z_4; z_1, z_2)$$

Assume that none of the ratios are 0 or ∞ . Then the **cross ratio (double ratio)** of these 4 points is:

$$(z_1, z_2; z_3, z_4) := \frac{(z_1; z_3, z_4)}{(z_2; z_3, z_4)}$$

If $(z_1, z_2; z_3, z_4) = \lambda$, then (for distinct z_1, z_2, z_3, z_4):

$$\begin{aligned} (z_3, z_4; z_1, z_2) &= (z_2, z_1; z_4, z_3) = (z_4, z_3; z_2, z_1) = \lambda \\ (z_3, z_1; z_2, z_4) &= \frac{1}{1 - \lambda}, (z_2, z_1; z_3, z_4) = \frac{1}{\lambda}, (z_2, z_3; z_1, z_4) = \frac{\lambda - 1}{\lambda} \\ (z_3, z_2; z_1, z_4) &= \frac{\lambda}{\lambda - 1}, (z_1, z_3; z_2, z_4) = 1 - \lambda \end{aligned}$$

Note that on the extended complex plane we can remove the conditions on simple ratios.

Claim 7.7 — $(z_1, z_2; z_3, z_4)$ is well-defined if $z_1, z_2, z_3, z_4 \in \overline{\mathbb{C}}$ and any three of them are distinct.

Proposition 7.8

Let $z_1, z_2, z_3, z_4 \in \overline{\mathbb{C}}$, any three of them are distinct. Then z_1, z_2, z_3, z_4 are cocyclic iff $(z_1, z_2; z_3, z_4) \in \mathbb{R} \cup \{\infty\}$.

Proof. Assume $z_1 = z_2$. Then we can check that $(z, z; z_3, z_4) \in \mathbb{R}$, therefore they always lie on a circle, which is true for any 3 points.

Now we assume that all points are distinct. Then:

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

If the cross ratio is real:

$$\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} = \overline{\left(\frac{z_1 - z_3}{z_1 - z_4}\right)} \cdot \overline{\left(\frac{z_2 - z_4}{z_2 - z_3}\right)}$$

Now after computations, we can rearrange the equation into a generalized circle equation with variable z_1 . This circle will contain points z_2, z_3, z_4 . If z_1 will satisfy the equation, that would mean that the cross ratio is real. $\square$

Proposition 7.9

Let $z_1, z_2, z_3, z_4 \in \overline{\mathbb{C}}$ be triple-distinct. Consider non-point circle $\mathcal{C}_0$ and inversion with respect to it. Then:

$$(z_1^*, z_2^*; z_3^*, z_4^*) = \overline{(z_1, z_2; z_3, z_4)} = (\bar{z}_1, \bar{z}_2; \bar{z}_3, \bar{z}_4)$$

Proof. $z^* = -\frac{C_0\bar{z}+D_0}{A_0\bar{z}+B_0}$. Then we can compute cross-ratio:

$$z_i^* - z_k^* = -\frac{(A_0D_0 - B_0C_0)(\bar{z}_i - \bar{z}_k)}{(A_0\bar{z}_i + B_0)(A_0\bar{z}_k + B_0)}$$

$$\frac{z_i^* - z_k^*}{z_j^* - z_k^*} = \frac{A_0\bar{z}_j + B_0}{A_0\bar{z}_i + B_0} \cdot \frac{\bar{z}_i - \bar{z}_k}{\bar{z}_j - \bar{z}_k}$$

$$(z_1^*, z_2^*; z_3^*, z_4^*) = \overline{(z_1, z_2; z_3, z_4)}$$

□

§8 Möbius transformations

Recall the formula of inversion: $z^* = -\frac{C_0\bar{z}+D_0}{A_0\bar{z}+B_0}$. We can consider more general functions and study them.

Definition 8.1. Suppose $a, b, c, d \in \mathbb{C}$. A **Möbius transformation** is a function of the type:

$$f : \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}, \quad f(z) = \frac{az + b}{cz + d}$$

Each Möbius transformation gives a 2x2 matrix $\mathcal{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Fact 8.2. A Möbius transformation is a constant map ($f(z) = z_0, \forall z$) iff the condition $\det(\mathcal{A}) = ad - bc = 0$ holds.

Proof. ($\Rightarrow$) :

$$\frac{az + b}{cz + d} = w \Rightarrow az + b = w(cz + d) \Rightarrow (a - cw)z + (d - sw) = 0 \Rightarrow a = cw, \quad b = dw$$

The proof of converse is similar, so we will omit it. $\square$

Claim 8.3 — If $\lambda \in \mathbb{C} \setminus \{0\}$, then $\mathcal{A}$ and $\lambda\mathcal{A}$ define the same Möbius transformations. The converse is also true, so if $\mathcal{A}_1$ and $\mathcal{A}_2$ define the same transformation $\Rightarrow$

$$\exists \lambda \in \mathbb{C}, \lambda \neq 0 : \lambda\mathcal{A}_1 = \mathcal{A}_2$$

Remark 8.4. A Möbius transformation is also called a **(projective) linear transformation**.

Let $z = \frac{z_1}{z_2}$, then $f(z_1, z_2) = \frac{az+b}{cz+d} = \frac{az_1+bz_2}{cz_1+dz_2}$. Now note that:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} az_1 + bz_2 \\ cz_1 + dz_2 \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

And if we consider a point $\frac{w_1}{w_2} = w$, it will be exactly the image of point z after a Möbius transformation. If we represent complex numbers in a projective way, we get that Möbius transformation is a linear transformation, because it has its own matrix.

Now we have one more interpretation of what complex number is: a complex number $z \in \overline{\mathbb{C}}$ is a equivalence class of pairs $(z_1, z_2) \in \mathbb{C}^2 \setminus \{(0, 0)\} : \frac{z_1}{z_2} = z$:

$$(\{(z_1, z_2) | z_1, z_2 \in \mathbb{C}\} \setminus \{(0, 0)\}) / \sim$$

The point ∞ naturally occurs in this set as equivalence class of pairs $(z, 0)$.

Remark 8.5. Complex plane defined this way (by equivalence classes) is called **projective complex line** and denoted $\mathbb{CP}^1$.

Then any Möbius transformation defines a linear function from $\mathbb{CP}^1$ to itself, saving the equivalence relation between elements.

Consider $m(z) = \frac{az+b}{cz+d}$ - a non-constant Möbius transformation. Then we can find z from its image $m(z)$ by finding a reverse matrix: $m^{-1}(z) = \frac{dz-b}{-cz+a}$. Then $m(z)$ is a bijective map on $\mathbb{CP}^1$. It is also a continuous map, and the inverse of this map is also a Möbius transformation.

Definition 8.6. We call function m a **homeomorphism**, if it is bijective and continuous, and its inverse is also continuous.

We can think of m as a kind of symmetry on $\overline{C}$, because it has 1-1 correspondence. Suppose $m_1(z)$ and $m_2(z)$ - 2 Mobius transformations. Then consider composition of these 2 functions. Because these 2 maps are linear, their composition will correspond to the pproduct of their matrices, therefore, will also be a Mobius transformation.

Proposition 8.7

A set of Mobius transformations is a group under the operation of composition of functions. It is easy to check that this set indeed is a group

Definition 8.8. A bijective function on X to itself is called **symmetry** of X .

Now let us consider a following decomposition of Mobius transformation:

$$w = \frac{az + b}{cz + d} = \frac{a}{c} + \frac{b - \frac{ad}{c}}{cz + d} = \frac{a}{c} - \frac{\delta}{c} \cdot \frac{1}{cz + d}$$

From the decomposition we can see that every Mobius transformation can be expressed as:

$$z \mapsto z + \frac{d}{c} \text{ (translation)} \mapsto cz + d \text{ (rotation+dilation)} \mapsto \frac{1}{cz + d} \text{ (reciprocation)}$$

1. **Translation** is a transformation of a form $z \mapsto z + z_0$, $z_0 \in \mathbb{C}$.
2. **Rotation** is a transformation of a form $z \mapsto ze^{i\phi}$, $\phi \in [0; 2\pi)$.
3. **Dilation** is a transformation of a form $z \mapsto Rz$.
4. **Reciprocation** is a tranformation of a form $z \mapsto \frac{1}{z}$.

Note that all of the transformations above can be expressed as composition of 2 inverses.

Proposition 8.9

Mobius tranformations are conformal. Mobius transformations carry real circles and lines to real circles and lines, and imaginary circles to inaginary ones. Moreover, the orientations of all circles are preserved iff they do not intersect the pole and for any 4 points their cross-ratio is preserved.

Proposition 8.10

Given three distinct points $z_1, z_2, z_3 \in \mathbb{CP}^1$ and another three points $w_1, w_2, w_3 \in \mathbb{CP}^1$, exists a unique Mobius transformation, taking z_i to w_i . To prove that, simply recall that Mobius transformations preserve cross ratios, so the image of any point will be determined uniquely.

Corollary 8.11

If function $w = f(z)$ from $\overline{C}$ to itself preserves cross-ratio, it is a Mobius transformation.

Remark 8.12. $\mathbb{CP}^1$ has a very large symmetry group. That means that for any 2 triplets of points exists a symmetry mapping this triplets one to another.

Corollary 8.13

Every real circle can be transformed into $|z|^2 = 1$ and every imaginary circle can be transformed into $|z|^2 = -1$ by a Mobius transformation.

Our goal is to classify Mobius transformations.

Definition 8.14. If m_1, m_2 are Mobius transformations, m_1 is **similar** to m_2 if exists a MT $I : m_2 = I^{-1}m_1I$. Note that this definition makes an equivalence relation on a set of all MT.

Remark 8.15. The set of all MT is NOT equal to the set of all matrices with non-zero determinants.

Definition 8.16. For a matrix A , its **trace** ($Tr(A)$) is equal to the sum of all elements on the main diagonal of the matrix.

We can introduce a function σ on a set of 2x2 matrices: $\sigma(A) = \frac{Tr(A)^2}{|A|-4}$. Then we will have a property $\sigma(\lambda A) = \sigma(A)$. Moreover, for every class of similar MT, the σ will be the same for all of them.

DUE TO SOME INCOVINIENCES, THE NOTES FOR THE FOLLOWING LECTURE ARE NOT PROVIDED.

Definition 8.17. A fixed point γ is called **attractive** for some function f if for all z in a certain neighbourhood of $\gamma \Rightarrow \lim_{n \rightarrow \infty} f^n(z) = \gamma$.

A fixed point γ is called **repulsive** if for any point z sequence $f^n(z)$ does not converge at γ .

If the sequence $f^n(z)$ does not approach any of the fixed points, then all the fixed points are called **indifferent**.

Example 8.18

$m(z) = kz, k \neq 0$. Then:

1. $|k| > 1$, then $m^l(z) = k^l z \rightarrow \infty$, so ∞ is an attracting FP.
2. $|k| = 1$, then $m^l(z) = k^l z$ so 0 and ∞ are indifferent.

The type of a fixed point does not change under the conjugation of MT.

Now we will think about iterating a function s times for arbitrary real s :

1. If $s \in \mathbb{N}$, iterating s times is defined
2. If $s \in \mathbb{Z}_-$, we can iterate f if it is invertible.

Now consider family of functions $\{F(s, z)\}_{s \in \mathbb{R}}$, satisfying the following condition:

- $F(0, z) = z$ - the identity
- $F(1, z) = f(z)$
- $F(s, F(r, z)) = F(s + r, z), \forall r, s \in \mathbb{R}$

Consider **Shroder's functional equation**:

$$k \cot \phi(z) = \phi(f(z))$$

and **Abel's functional equation**:

$$\psi(z) + 1 = \psi(f(z))$$

where f is given, and ϕ or ψ is an unknown function.

Then we get that a non-parabolic MT is a member of equivalence class of a function $m(z) = kz$ for some k .

Now suppose $\gamma_{1,2}$ are fixed points for some MT m . Then we can write m as follows:

$$w = m(z) = \frac{az + b}{cz + d} = \frac{(k\gamma_2 - \gamma_1)z + (1 - k)\gamma_1\gamma_2}{(k - 1)z + \gamma_2 - k\gamma_1}$$

§9 Hyperbolic Geometry

Klein Enlarger Programme: Let us consider set $\mathcal{M}$ - set of all non-constant MT. Then can we define geometry using its subgroups? Let ϵ be the set of all rigid motions. Euclidean geometry is the study of objects which are invariant under ϵ , so if we change ϵ to something different, we will construct some kind of geometry. Let us consider

$$\mathcal{U} = \{m(z) \mid m(\{|z| = 1\}) = \{|z| = 1\}\}$$

a subgroup of MT, which does not change the unit circle. Geometry base on $\mathcal{U}$ is called **hyperbolic geometry**.

$m(z) \in \mathcal{U}$ must send a pair of points symmetric with respect to unit circle (S^1) to another pair of such points. Then we have:

$$\frac{a}{d} = \frac{b}{c} = \frac{c}{b} = \frac{d}{a} = q \in \mathbb{C}^*$$

So we have $|q|^2 = 1 = e^{i2\alpha}$. Then $d = e^{i2\alpha}\bar{a}$ and $c = e^{i2\alpha}\bar{b}$. After choosing an a , we will have MT $m(z) = \frac{az+b}{\bar{a}z+\bar{b}}$. Now we have a clear representation of $\mathcal{U}$:

$$\mathcal{U} = \{m(z) = \frac{az+b}{\bar{a}z+\bar{b}} \mid a, b \in \mathbb{C}, |a| \neq |b|\}$$

Let us denote the region inside S^1 as C_+ and outside - C_- . Because m is continuous, $m(C_\pm) = C_\pm$. If $m(C_+) = C_+$:

$$m(0) = \frac{b}{\bar{a}} \in C_+ \Leftrightarrow \left|\frac{b}{\bar{a}}\right| < 1 \Leftrightarrow |b| < |a|$$

$$\mathcal{U}_+ := \{m \in \mathcal{U} \mid m(C_+) = C_+\} \subset \mathcal{U} \subset \mathcal{M}$$

If $a = 0$, then $bc \neq 0$. Using similar arguments for b , we get $|q| = 1$.

Take $m(z) = \frac{az+b}{bz+\bar{a}} \in \mathcal{U}_+$. Then:

1. $a = 0 \Rightarrow m(z) = \frac{b}{bz}$ is a **hyperbolic involution**.

2. $a \neq 0$. Then $m^{-1}(0) = -\frac{b}{\bar{a}} := z_1$. Set $a = re^{i\alpha}$, then $m(z) = \frac{a}{\bar{a}} \cdot \frac{z-z_1}{1-\bar{z}_1z} = e^{2i\alpha}m_{z_1}(z)$.

Now let us consider $\sigma(m) = \frac{4}{1-|z_1|^2} - 4 > 0$. Let m_{z_1} be MT which takes point z_1 to 0. Its fixed points will be $\pm e^{i\phi_1}$. For a MT we can compute its **characteristic** constant $k = \frac{1+r_1}{1-r_1}$.

So now we can iterate $m_{z_1}^s(z) = \frac{k^s-1}{k^s+1}e^{i\phi_1}$, $s \in \mathbb{R}$. Then $s \rightarrow \pm\infty \Rightarrow m^s \rightarrow \pm e^{i\phi_1}$. By conjugation and rotation:

$$e^{it}m_{z_1}^s(e^{-it}z) = \frac{z - e^{it}z_s}{1 - e^{-it}\bar{z}_sz}$$

Proposition 9.1

Every element in $\mathcal{U}_+$ can be expressed as

$$m(z) = R_1^{t_1} \cdot m_{\frac{1}{2}}^s \cdot R_1^{t_2}$$

where $R_1 = \begin{pmatrix} e^i & 0 \\ 0 & 1 \end{pmatrix}$.

$\mathcal{R} = \{\text{MT}m \mid m((z, -\frac{1}{\bar{z}})) \mapsto (w, -\frac{1}{\bar{w}})\}$ - group of rotational MT.

ϵ is the group of all rigid motions in $\mathbb{C} \cong \mathbb{R}^2$.

Then $\mathcal{U}_+ \cup \mathcal{R} \cup \epsilon \in \mathcal{M}$. They produce **hyperbolic, spherical and euclidian** geometries.

So what is exactly a Euclidean geometry? Suppose we have a set of figures which remain the same under some sort of transformations. This geometry is the study of properties of figures which remain unchanged under those transformations. Then to generalize the concept of geometry, we can choose a transformation and a set of figures constant under it, and study the properties of those figures with respect to those transformations.

Suppose $\epsilon = \{az + b \mid |a| = 1\}$.

1. $a = 1 \Rightarrow z \mapsto z + b$ - translation (parabolic).

$$T_b(z) = z + b; \quad T_b^s(z) = z + sb, \quad b \in \mathbb{C}, \quad s \in \mathbb{R}$$

When the z, b are fixed, changing the value of s will produce a line.

2. $a \neq 1 \Rightarrow \exists$ fixed point $\gamma = \frac{b}{1-a}$. $a = e^{i\phi}$, $m(z) = az + b = \gamma + e^{i\phi}(z - \gamma)$, $m^s(z) = \gamma + e^{is\phi}(z - \gamma)$. This transformation will produce a unit circle with center at γ .
3. In $\mathbb{R}^2 \cong \mathbb{C}$ we have a notion of distance. Distance is a 2 point function which is invariant under ϵ .

Definition 9.2. A **$\mathcal{G}$ -geometry** is a subgroup $\mathcal{G} \in \mathcal{M}$ together with a domain $\mathcal{D} \in \overline{\mathcal{C}}$, where $\mathcal{D}$ is $\mathcal{G}$ -invariant.

Definition 9.3. 1. A **$\mathcal{G}$ -circle** is a trajectory of an elliptic MT.

2. A **$\mathcal{G}$ -hypercycle** is a trajectory of a proper hyperbolic MT.
3. A **$\mathcal{G}$ -horocycle** is a trajectory of a parabolic MT.
4. A **$\mathcal{G}$ -lines** is either $\mathcal{G}$ -circles or $\mathcal{G}$ -cycles which lie in the straight line in ordinary Euclidean plane.

Example 9.4

$\mathcal{G} = \mathcal{U}_+$, $\mathcal{D} = \{z \in \mathbb{C} \mid |z| < 1\}$. Then $\mathcal{D}$ is $\mathcal{G}$ -invariant. The domain $\mathcal{D}$ is sometimes called the **hyperbolic disk**. There are 2 properties of $\mathcal{U}_+$ acting on $\mathcal{D}$:

1. $\mathcal{U}_+$ acts transitively, but not 2-fold transitively, so for 2 arbitrary points it can carry 1 to the other, but it cannot carry any 2-tuple to the arbitrary other.
2. The stabilizer subgroup consists of rotations.

These 2 statements imply the existence of non-constant 2 point function. We fix $z_1, z_2 \in \mathcal{D}$, $m \in \mathcal{U}_+$. Consider $m_{z_1}, m_{z_2}, m_0 := m_{w_2} \cdot m \cdot m_{z_2}^{-1}$, $0 \mapsto 0$. Then m_0 must be a rotation, so $f(z_1, z_2) := |m_{z_2}(z_1)| = |\frac{z_1 - z_2}{1 - \bar{z}_2 z_1}|$. This is an attempt to define the metric in this geometry, but is it satisfying all the properties required?

Question 9.5. Does this function satisfy the triangle inequality?